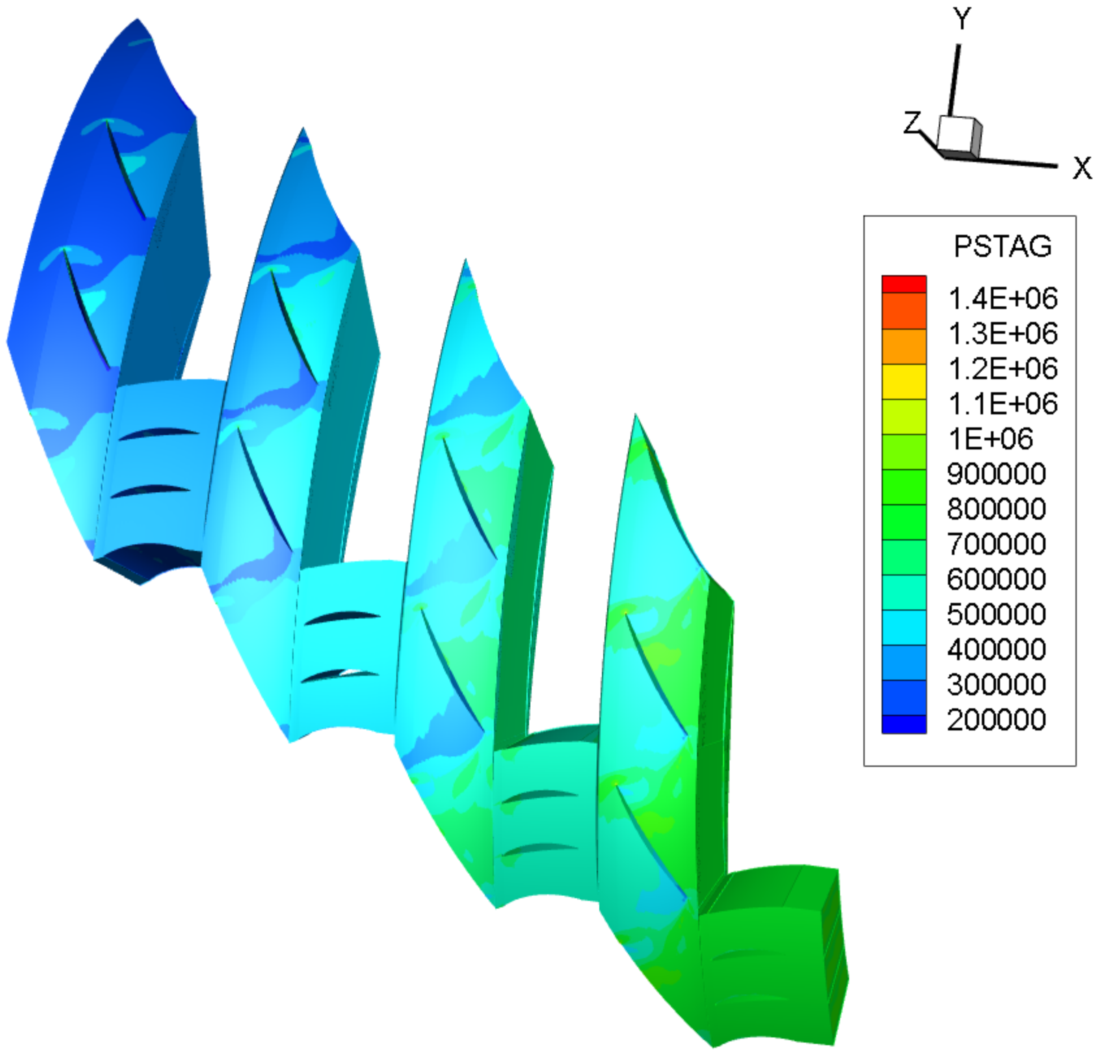
Delft University of Technology

AE4206 Turbomachinery



Assignment software tutorial

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February 29, 2020

1 Introduction

This tutorial contains some basic instructions on how to use Meangen, Stagen and Multall to analyze the performance of an axial compressor. Many of the instructions listed in this document can however also be used for different machine types. In this project, it's required to perform a CFD analysis of the first stage of a compressor. In Figure 1.1, an example workflow process can be seen on how the teams are to perform this analysis and end up with a satisfactory design.

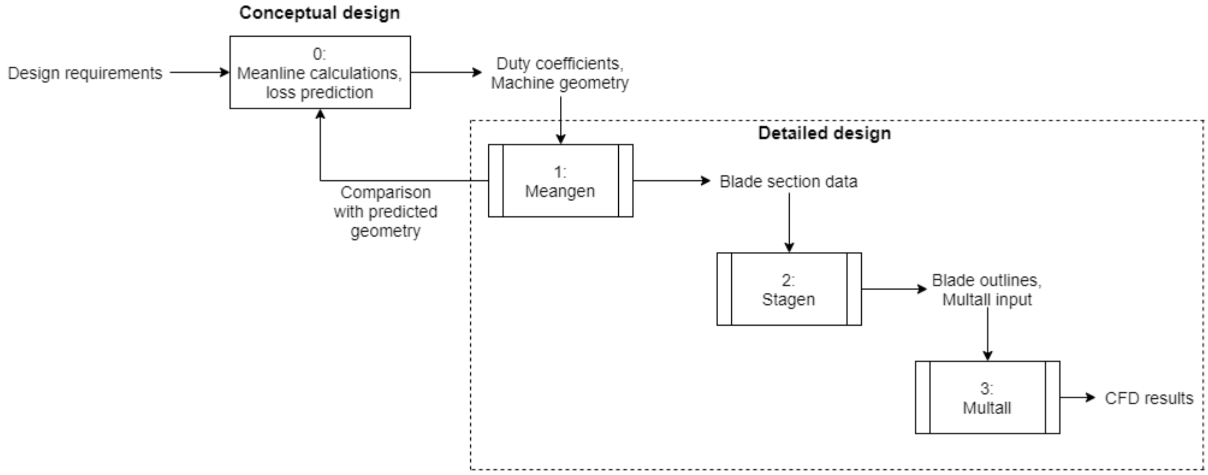

Figure 1.1: Workflow process of performing the CFD analysis

In block 0, the conceptual design process takes place, where based on the compressor requirements, you calculate the desired set of duty coefficients and geometry of the compressor by performing meanline calculations and predicting the losses with knowledge obtained in the lectures and other sources. Once the preliminary design has been chosen, the detailed design phase can start, in which the CFD analysis takes place.
The first step is by using a program called Meangen. Meangen generates the compressor geometry based on the input resulting from the preliminary design. You can compare this geometry with the one you predicted during the conceptual design phase to verify your approach. Meangen puts out more detailed data on the metal angles of the blades and thickness distribution.
In block 2, called Stagen, input from Meangen is used to generate the outline of the blades, which can then be inspected using TecPlot. Stagen also provides the input data for the CFD analysis, including the mesh, maximum number of iterations and convergence criteria. The CFD analysis itself is then initiated in block 3 by a program called Multall. The outputs from multall include the total mass flow, machine power and the flow field itself. The latter can be inspected in detail using TecPlot.

2 Using Meangen

In this chapter, the first steps of the machine analysis are described. Before being able to do a detailed analysis of the flow within the machine, you of course need a machine geometry. This is what Meangen is used for. Meangen is essentially a **meanline design program**. From a set of duty coefficients, mass flow rate, number of stages and the stage power, Meangen constructs the meridional flow path of the machine. Note that Meangen **does not apply any loss model**, so you'll have to calculate the stage loss coefficients yourself in order to predict the machine efficiency. It's important you do this before doing the detailed analysis, as the flow channel produced by meangen will probably be too narrow for the design mass flow when assuming an isentropic machine.
For Meangen to function, you will need the Meangen.exe executable and the input file Meangen.IN. Meangen.IN contains all the instructions and data Meangen needs to construct the meridional flow path. Once you have downloaded the executables, you can use them by opening a command window, navigating to your destination folder(or do this using *system* and *sprintf* in Matlab for example) and using the following command.

$$\rightarrow \text{path to executable/Meangen17.4.exe} \tag{2.1}$$

Meangen then asks whether the instructions should be collected from an input file or are to be filled in on the screen. If you have an input file prepared and in the current directory, you can type "F" and Meangen will generate the design automatically. If you want to follow the same design approach as given in the example file, you can use this method right from the start. In case you like to take a different approach/analyze more stages, it might be more beneficial to type "S" and fill in your design parameters using the command window. Meangen generates a meangen.OUT file, in which all instructions you put in are listed, which you can use as an input file for your next design. In Figure 2.1, the meangen.OUT file from the design example can be seen. The design inputs are on the left and the description of the parameter on the right. Next up follows a step-by-step description of the Meangen inputs.

```
C                               TURBO_TYP,"C" FOR A COMPRESSOR,"T" FOR A TURBINE
AXI                             FLO_TYP FOR AXIAL OR MIXED FLOW MACHINE
    287.500      1.400          GAS PROPERTOES, RGAS, GAMMA
      2.500    340.000          POIN,  TOIN
     1                          NUMBER OF STAGES IN THE MACHINE
M                               CHOICE OF DESIGN POINT RADIUS, HUB, MID or TIP
    13700.000                   ROTATION SPEED, RPM
        88.000                  MASS FLOW RATE, FLOWIN.
A                               INTYPE, TO CHOOSE THE METHOD OF DEFINING THE VELOCITY TRIANGLES
   0.800  0.800  0.260          REACTION, FLOW COEFF., LOADING COEFF.
A                               RADTYPE, TO CHOOSE THE DESIGN POINT RADIUS
         0.280                  THE DESIGN POINT RADIUS
         0.050          0.040   BLADE AXIAL CHORDS IN METRES.
         0.250          0.500   ROW GAP  AND STAGE GAP
    0.00000    0.00000          BLOCKAGE FACTORS, FBLOCK_LE,  FBLOCK_TE
         0.900                  GUESS OF THE STAGE ISENTROPIC EFFICIENCY
    5.000     5.000             ESTIMATE OF THE FIRST AND SECOND ROW DEVIATION ANGLES
   -2.000  -2.000               FIRST AND SECOND ROW INCIDENCE ANGLES
    0.50000                     BLADE TWIST OPTION, FRAC_TWIST
N                               BLADE ROTATION OPTION , Y or N
   88.000  92.000               QO ANGLES AT LE  AND TE OF ROW 1
   92.000  88.000               QO ANGLES AT LE  AND TE OF ROW 2
N                               DO YOU WANT TO CHANGE THE ANGLES FOR THIS STAGE ? "Y" or "N"
Y                               IS OUTPUT REQUESTED FOR ALL BLADE ROWS ?
N     ROTOR No.   1 SET ANSTK = "Y" TO USE THE SAME  BLADE SECTIONS AS THE LAST STAGE
   0.0750  0.4000               MAX THICKNESS AND ITS LOCATION FOR ROTOR   1 SECTION No.  1
   0.0750  0.4000               MAX THICKNESS AND ITS LOCATION FOR ROTOR   1 SECTION No.  2
   0.0750  0.4000               MAX THICKNESS AND ITS LOCATION FOR ROTOR   1 SECTION No.  3
N     STATOR No.  1 SET ANSTK = "Y" TO USE THE SAME  BLADE SECTIONS AS THE LAST STAGE
   0.1000  0.4500               MAX THICKNESS AND ITS LOCATION FOR STATOR  1 SECTION No.  1
   0.1000  0.4500               MAX THICKNESS AND ITS LOCATION FOR STATOR  1 SECTION No.  2
   0.1000  0.4500               MAX THICKNESS AND ITS LOCATION FOR STATOR  1 SECTION No.  3
```

Figure 2.1: meangen.OUT file from the example

1. On the first line, it asks for the machine type, being a compressor("C") or a turbine("T"). The second line specifies whether this is an axial("AXI") or mixed flow("MIX") machine.

2. The next two lines ask for the gas properties(gas constant and specific heat ratio) of the working fluid and the inlet total pressure in bar and the inlet total temperature in Kelvin.

3. On line 5, you specify the number of stages to be produced by Meangen.

4. On line 6, Meangen asks for hub, mid or tip based design. This means that Meangen will keep the given radius constant over the stage. So in this case(”M”), meangen will keep the mean radius of the rotor and stator of this stage constant.

5. The next two lines ask for the rotation speed and machine mass flow rate in $\mathrm{kg\,s^{-1}}$.

6. Next, Meangen asks for how you’d like to define the velocity triangles of this stage. With option ”A”, you insert the degree of reaction, flow coefficient and loading coefficient. Option ”B” allows you to specify the flow coefficient, stator exit angle and rotor exit angle(relative). With option ”C”, you specify the flow coefficient and relative rotor inlet and exit angle. Finally, option ”D” asks for stage reaction and first blade row inlet and exit angle.

7. The next part has to do with the calculation of the machine design radius. This can be done by inserting the design radius directly(option ”A”) or by inserting the stage enthalpy change in $\mathrm{kJ\,kg^{-1}}$(option ”B”).

8. On lines 13 and 14, the axial chord lengths and row and stage gaps are specified. The first value for axial chord represents the first blade row chord and the second value the second blade row. After this, the row gap(space between the rotor and stator) and stage gap(space between this and following stage) are specified. These are taken as a fraction of the stage first blade row chord(so in this case the rotor chord, resulting in a row gap of 0.01 m and a stage gap of 0.02 m).

9. Next, Meangen asks for the blockage factors at the first blade row leading edge and second blade row trailing edge. The blockage factor is the ratio between the local boundary layer displacement thickness and the blade height. Meangen uses these values to scale up the machine sufficiently in order to allow for the required mass flow to flow through.

10. On the next line, Meangen asks for a guess of the isentropic efficiency of the stage. Meangen will use this to calculate the pressure at the end of the stage and size the exit blade height appropriately.

11. The next two lines have to do with the incidence and deviation angles of the blade rows. On the first line, the deviation angles of the rotor and stator rows are to be filled in and on the second line the incidence angles. In Figure 2.2, once can see the definition of the incidence and deviation angles. Meangen will add these angles to the inflow/outflow angles resulting from the set of duty coefficients of the blade row in order to produce the blade metal angles.

12. On line 19 of Figure 2.1, the blade twist option is specified. This has to do with the level of vortex-free design of the stage. Filling in 1.0 results in the blade to yield full vortex-free design and 0.0 leads to a prismatic blade. The user can also fill in a value in between 1.0 and 0.0 in order to for instance go for a controlled-vortex design. The mathematical law for this is

$$\beta = (1-T)\beta_{\text{design}} + T tan^{-1}\left(tan(\beta_{\text{design}})\frac{r_{\text{design}}}{r} + \frac{1}{\phi(r)}\left[\frac{r_{\text{design}}}{r} - \frac{r}{r_{\text{design}}}\right]\right) \tag{2.2}$$

Here, β is the relative blade angle (or absolute in case of a stator), T the fractional twist input value, β_{design} the blade angle of the design section(depending on your input for hub, mean or tip design). r_{design} is the radius of the design section and r the radius of the local section. $phi(r)$ is the local flow coefficient, equal to $\phi_{\text{design}}\frac{r_{\text{design}}}{r}$ in case axial velocity is assumed constant over the blade span. In Figure 2.4, the blade sections at the hub, mid and tip of a turbine stage can be seen for twist values of 0.5 and

13. For the next input, the user gets the option to change the blade angles later in case the resulting stage geometry is not as desired. This is only really useful when using the screen approach when using Meangen, as the user gets updated on the machine design after filling in certain design parameters. When using an input file for the Meangen commands, it's advised to just fill in "N" at this option.

14. The next input has to do with the taper and sweep of the blade rows. The user is asked to fill in the QO angles at the leading and trailing edge of the first and second blade row. In Figure 2.3, an illustration is provided of the meaning of these angles, where Q1 and Q2 are the respective leading and trailing edge QO angles. If the user were to fill in 90 degrees for all QO angles, both blade rows would consist of blades with a taper ratio of 1.0.

15. The next input is again a choice for the user to change the flow angles for the current stage and is only relevant when filling in the design parameters on screen.

16. The next option on line 24 is important to write "Y", as it will write the Stagen files for all blade rows, which will have to be used later during the CFD analysis.

17. The last two inputs have to do with the thickness distribution over the blade surface. By filling in "N" on lines 25 and 29, the user gets the option to change the thickness-to-chord ratio and chordwise location of maximum thickness on the blade on the root(section 1), mid(section 2) and tip(section 3) of the blade row. The first column represents the thickness-to-chord ratio and the second column the location of maximum thickness.

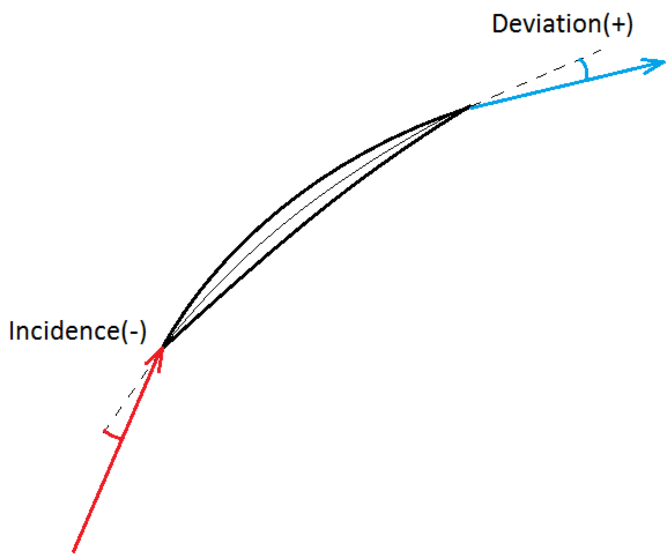

Figure 2.2: Blade incidence and deviation angle definition

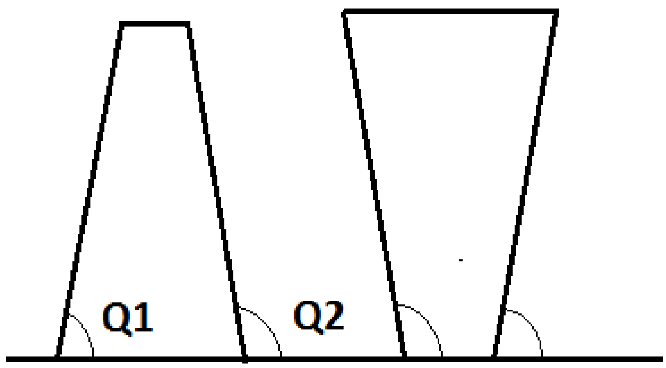

Figure 2.3: Blade QO line angles

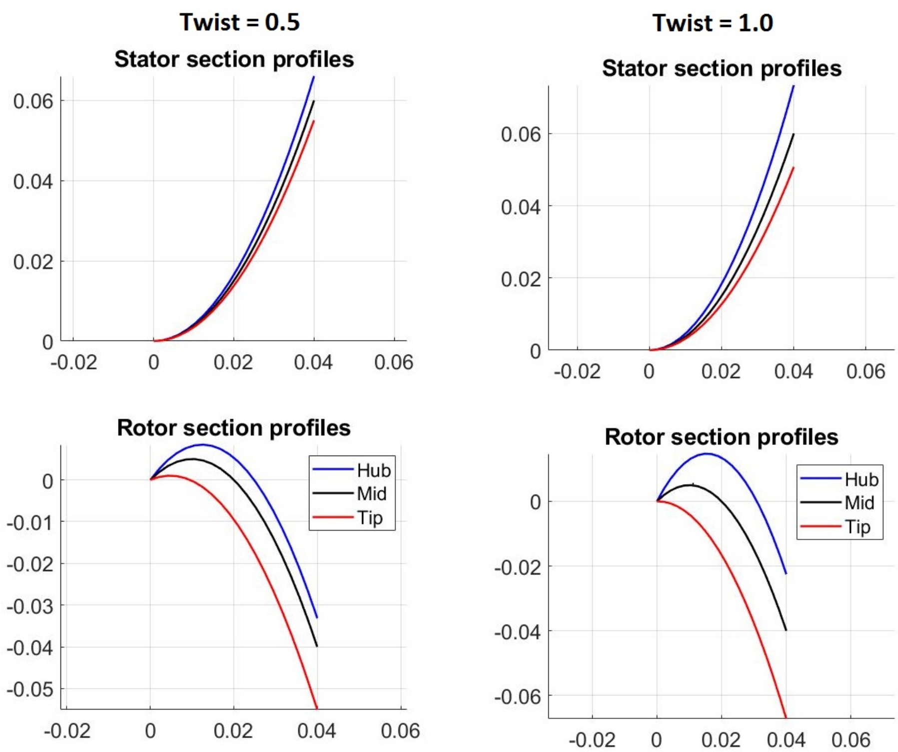

Figure 2.4: Spanwise profile variation with twist value

After filling in all design parameters and initiating Meangen, a meangen.OUT, meandesign.OUT and stagen.dat file are created. In meangen.OUT, all commands you inserted can be found as how they were interpreted by Meangen. If there are differences between the meangen.OUT and the meangen.IN file you used, you probably inserted a command the wrong way. meandesign.OUT contains a more detailed description of the machine geometry, flow deflection angles and flow properties along the mean line. stagen.dat contains the information on the blade profiles, which can be loaded into TecPlot for inspection.

3 Using Stagen

In this chapter, a brief tutorial is given on how to use Stagen and how to use it to adjust the convergence criteria and iteration count for Multall. After using Meangen to generate a meanline design and preliminary geometry, Stagen is used to generate a suitable mesh for the CFD analysis in Multall. This is done by opening a command window, navigating to your current directory(where you have stored the output files from Meangen) and typing the following command:

$$\rightarrow \text{path to exectuable/Stagen17.2.2.exe} \tag{3.1}$$

The message will then appear, asking for whether the Stagen input file is called "stagen.dat". This is the case by default, so after you type "Y", Stagen will read the stagen.dat file generated by Meangen earlier.
Stagen then produces 4 files. *stage_new.dat* contains all inputs Multall requires to function. This file has to be used for subsequent steps. Stagen also creates a file named *stage_old.dat*. This input file is compatible with older versions of Multall, but is not compatible with the version you'll be using during the project. *blade_profiles.tec* contains the blade profiles in a format which can be read by Tecplot. *stagen.out* lists the blade profile coordinates, in case you'd like to plot them using a program other than Tecplot.
In order to inspect the blade profiles in Tecplot, download and install Tecplot 360 from the student software resources portal, open it and load the *blade_profiles.tec* file. On the left bar, click on the "Mapping Style" button to select the blade profiles you want to inspect.

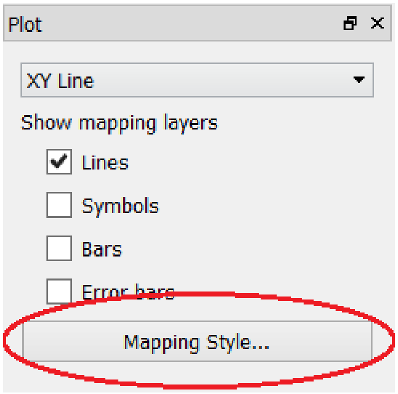

Figure 3.1

The following window then appears, in which you can select which curves you want to see(YUP is upper curve, YLOW lower curve and THICK the blade thickness) and which blade and section under the ZONE tab(section 1, 2 and 3 are hub, mean and shroud sections respectively).

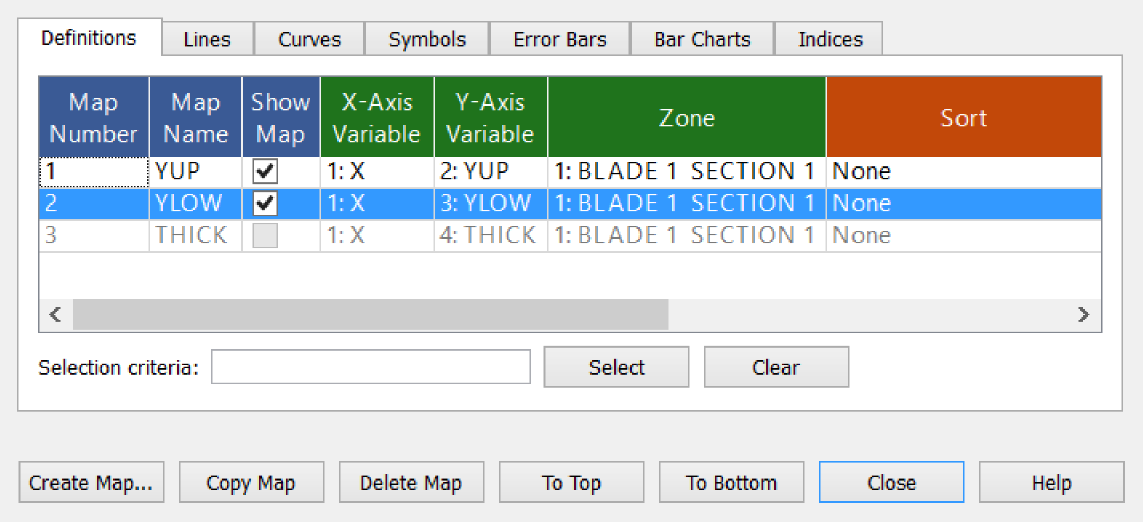

Figure 3.2

The plotted blade profile should look something like the one shown in Figure 3.3, where the red line, green line and blue line are the upper, lower and thickness curves respectively.

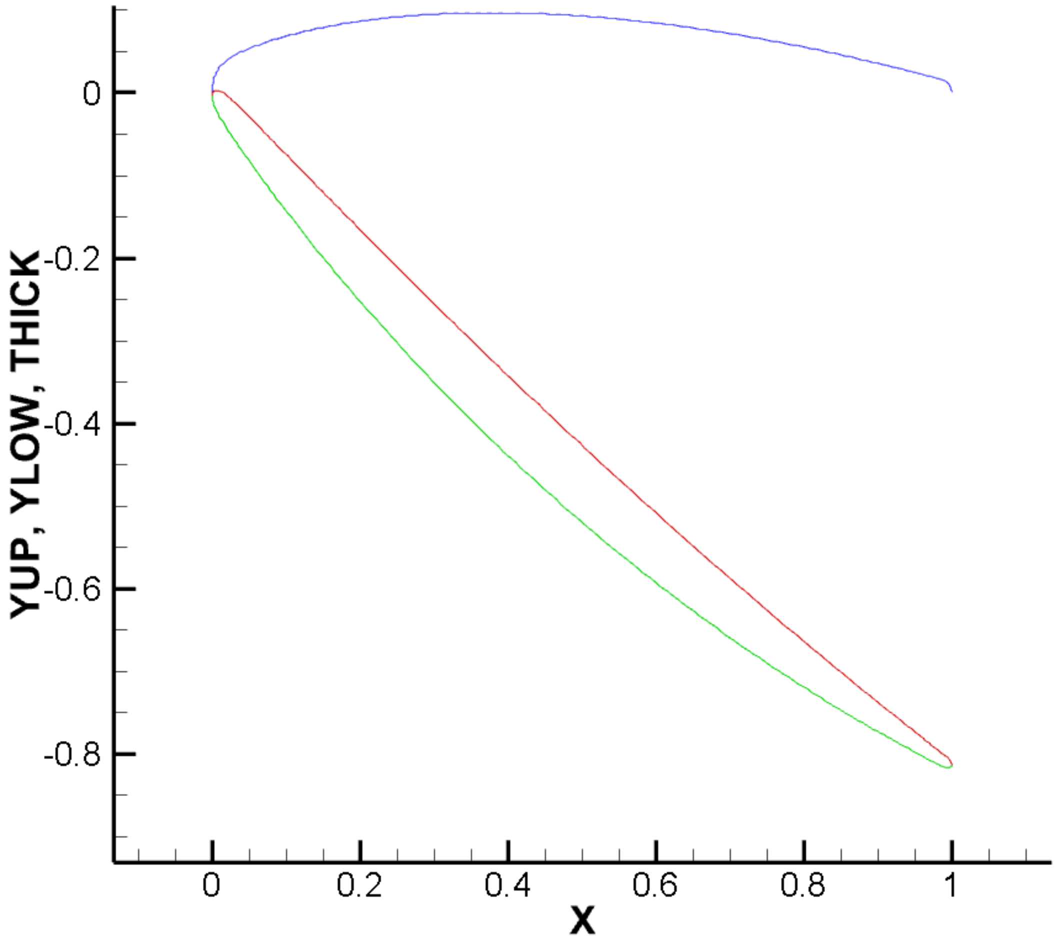

Figure 3.3

During detailed design, you might want to adjust certain aspects of the blade shape to increase performance. These can be adjusted in *stagen.dat* for each blade and its respective sections. In

Figure 3.4, the line defining the blade profile specifications for section 1 of the rotor blade row is highlighted.

```
*********************** ROW NUMBER 1 ************************************
**********STARTING NEW BLADE SECTION, SECTION NUMBER 1 **************
*************************BLANK LINE*****************************************
   1                      INTYPE- TYPE OF BLADE GEOMETRY INPUT
   6  200    4            NPIN, NXPTS, NSMOOTH
     0.0000    -48.7450 BLADE CENTRE LINE ANGLES
     0.1139    -45.7533 BLADE CENTRE LINE ANGLES
     0.2903    -42.4028 BLADE CENTRE LINE ANGLES
     0.5018    -38.6522 BLADE CENTRE LINE ANGLES
     0.7399    -34.4638 BLADE CENTRE LINE ANGLES
     1.0000    -29.8092 BLADE CENTRE LINE ANGLES
   0.0200    0.0100    0.0750    0.4000    0.0200    0.0100    2.0000    BLADE PROFILE SPECIFICATION
   1.0000    0.0000    1.0000                        FCHORD, FPERP, FTKSCALE
   0.0000    0.5000    0.5000                        ROTN,XROT,YROT
   0.5000    0.2500  -48.7450  -29.8092              XCUP, XCDWN, BETUP, BETDWN
 BLANK LINE
   9                 NUMBER OF POINTS ON THE STREAM SURFACE.
  -0.050000   -0.025805   -0.001775    0.051313    0.057495    0.063741    0.101316    0.121905
   0.142500
   0.234036    0.234165    0.235325    0.243956    0.244334    0.244449    0.245709    0.245909
   0.245931
  -0.001775    0.051313    0.235325    0.243956  LEADING AND TRAILING EDGE COORDINATES
   1.0000    0.0000    0.0000    0.0000    0.000 FCENTROID, FTANG, FLEAN, FSWEEP, FAXIAL
   1.0000    0.0000                              FSCALE, FCONST
```

Figure 3.4: Section from *stagen.dat*, highlighting the line defining blade profile specifications

The first two numbers represent the leading edge and trailing edge thickness ratio($\frac{t}{c}$). The third and fourth numbers represent the maximum thickness-to-chord ratio and its chordwise location respectively. These values were defined at the last couple of commands in Meangen, but can be changed here if desired. The two numbers after those represent the length of the leading and trailing edge. High numbers translate to a more sharp edge, while low numbers result in more rounded edges. This is illustrated in Figure 3.5, where on the left, 0.1 is used as the leading edge length, while on the right 0.01 was used.

Figure 3.5: Comparison between high and low leading edge length parameter

The final number has to do with the flatness of the thickness profile. High numbers result in a very flat profile, where the blade thickness retains its maximum thickness for most of the blade length, while low numbers result in a more diamond-type shape. In Figure 3.6, the effect of changing the flatness parameter is shown. On the left, a blade profile with a high number(5.0) is shown and on the left a blade with a low number(1.0).

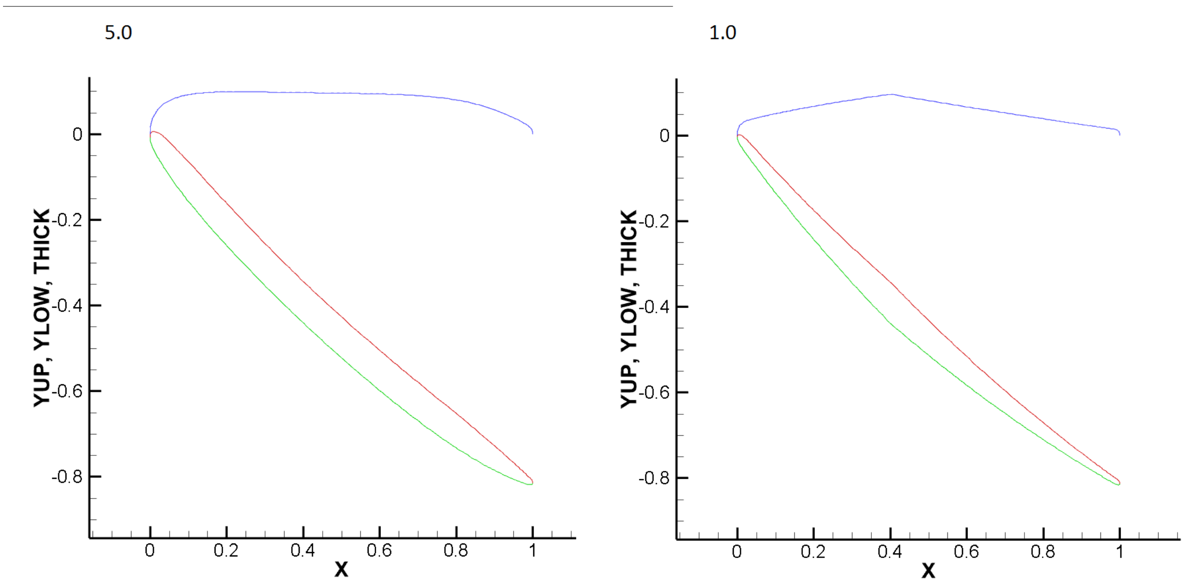

Figure 3.6: Comparison between high and low flatness parameter

There are more parameters which can be adjusted in the subsequent lines, but the ones shown in this chapter were deemed the most influential. Before starting the CFD analysis in Multall, you can adjust the maximum number of iterations and the convergence criterion. Unfortunately, this isn't as straightforward as changing the blade shape. In order to adjust these parameters, one has to adjust the source code of Stagen itself. When opening the source code in for instance Notepad++, the Multall control variables can be found between line 166 and 232. NMAX on line 170 represents the maximum number of iterations. Once this iteration count is reached, Multall will finalize its calculations and write its results, irrespective of the current convergence. The convergence limit is called CLIM and can be found on line 227. When the convergence factor gets below this number, Multall will again put out its results.
Stagen is written in Fortran and in order to compile it into a functional executable, you need a Fortran compiler such as G95. Once you have this program installed, open a command window, navigate to the folder where the source code is kept and then type *G95 stagen-17.2.f*. The new executable is then created in the current folder. By then replacing the Stagen executable in your executable folder with the new one, Stagen will put out the adjusted settings as input for Multall.

4 Using Multall and Tecplot for post-processing

After Stagen is done, the CFD analysis in Multall can start. In order to initiate Multall, open a command window, navigate to the folder where *stage_new.dat* is stored and type the following command:

>>path_to_executable/Multall17.5.exe <stage_new.dat >results.txt (4.1)

The last part of the command transfers all result data to the file *results.txt*. You can also leave this out. In that case, all the outputs from Multall will be displayed in the command window. By reloading the *results.out*, you can check on progress. Solving the flow field can take a long time, so don't be surprised if it takes half an hour or more!

In case you'd like Multall to stop and put out the results up to now, open the *stopit* file, which was made when initiating Multall in your current folder. Open this file in a text editor and replace the 0 with a 1. Multall will read this and terminate the solving process. To inspect the flow field and extract important flow parameters, you will have to convert the field data to a format readable by Tecplot. This can be done by executing the following command in the command window.

>>path_to_executable/convert2tecplot.exe (4.2)

The program will ask how many blade passages you want to show. 3 or 4 should be enough to properly show the flow field. Any higher will generate huge data files and take forever to load. After the data is converted, open Tecplot and load the *tecplot-input.dat* file. This will again take a few minutes to load. After Tecplot has finished loading, tic the "Contour" box on the left to show the flow field. This can be seen in Figure 4.1.

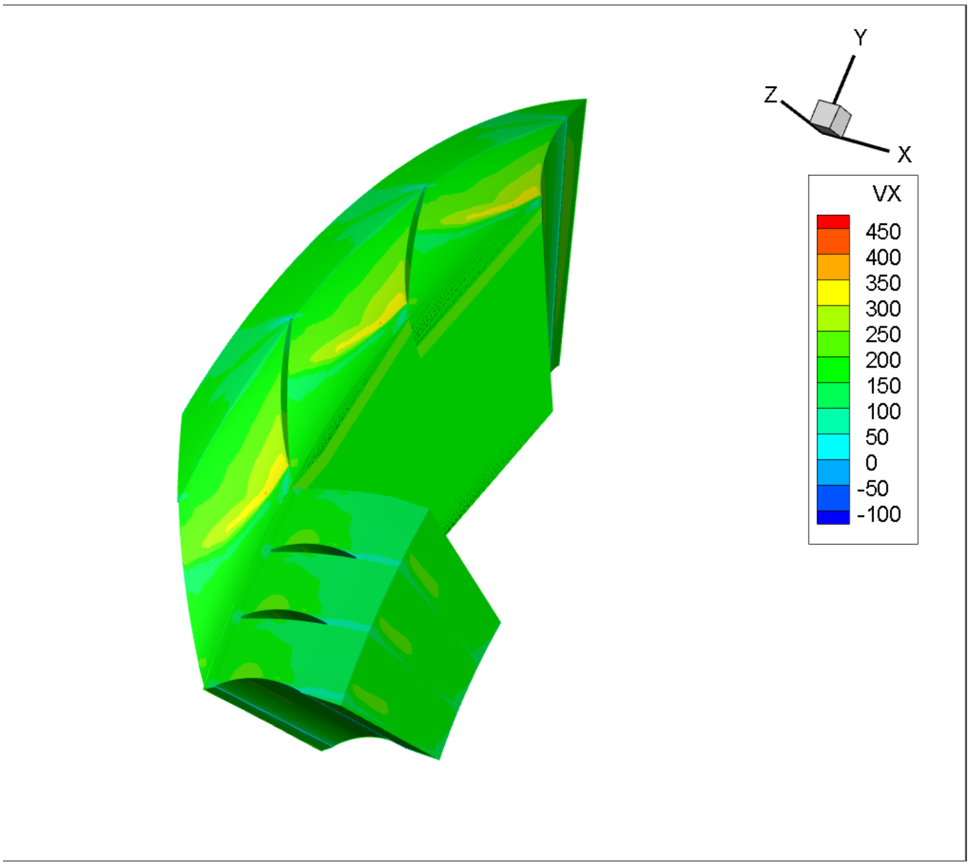

Figure 4.1: Axial velocity contour of the first stage

You can show different flow properties by clicking on "Details" behind the Contour box. You can visualize slices of the flow field by selecting "Slices" and selecting the flow property you want to show under "Contour". Below, the K-slice 19 with stagnation pressure contours.

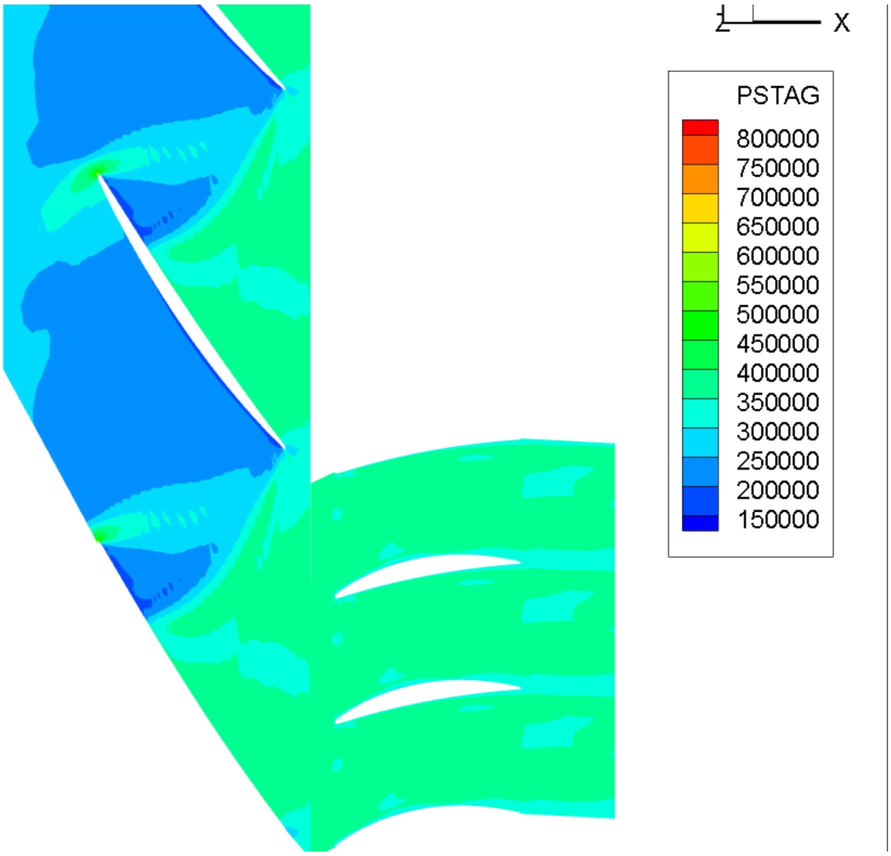

Figure 4.2: Blade plane showing stagnation pressure contours

Within Tecplot, you can define variables, equations and perform integrals, which you can use to distinguish 2D and 3D losses or compute efficiencies.